

Executive Summary: Regression Analysis

TikTok claims classification project

OVERVIEW

The TikTok data team aims to develop a machine learning model to classify user submissions as claims or opinions. Earlier analysis showed that verified users are more likely to post opinions. Since the ultimate goal is to distinguish claims from opinions, it is important to understand and predict the behavior of verified accounts. To support this, the data team built a logistic regression model to predict `verified_status`.

PROJECT STATUS

The variable `verified_status` was chosen for this regression model due to the observed relationship between account verification and video content. Logistic regression was selected because it is well-suited to the data type and distribution.

Model Results

The logistic regression model achieved a weighted precision of 69% and a weighted recall of 66%, resulting in an F1 score of 66%. These results provide insights into the relationship between video features and verified status, which are discussed further in the “Key Insights” section.

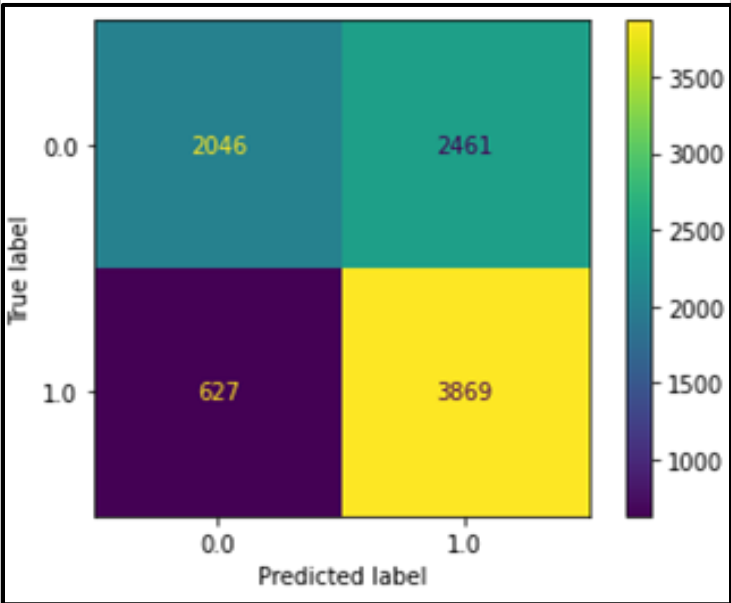
NEXT STEPS

The next step is to build a classification model to predict the claim status of user videos. This represents the final objective of the project as originally defined by the TikTok team. With the insights gained from analyzing user behavior, there is now sufficient context to interpret the results of this model effectively.

KEY INSIGHTS

The logistic regression model indicates that longer videos are associated with higher odds of the user being verified. Other video features have small estimated coefficients, suggesting they have little association with verified status. Consequently, aside from video length, the other features do not appear to meaningfully influence whether a user is verified.

Confusion matrix for logistic regression model



Upper-left: Number of videos from unverified accounts correctly classified (true negatives).

Upper-right: Number of videos from unverified accounts misclassified (false positives).

Lower-left: Number of videos from verified accounts misclassified (false negatives).

Lower-right: Number of videos from verified accounts correctly classified (true positives).